



STRATHMORE UNIVERSITY

iLabAfrica Research Centre

Nairobi, Kenya

UBUNTUNET ALLIANCE: CALL FOR PROPOSALS

AI for Climate Resilience in Africa

Co-funded by the European Union through AfricaConnect 4

Predict. Prioritise. Protect:

An AI-Enabled Compound Climate Disruption Decision-Support System for Kenya's 47 Counties

Predicting and Prioritising Household and County-Level Climate Risk
from Water, Electricity, Road and Flooding Shocks using Machine Learning,
Explainable AI and Data Envelopment Analysis

Full Proposal: Grant Application

Lead Institution:	Strathmore University, iLabAfrica Research Centre
Principal Investigator:	David Mauti
Technical Lead:	Jakes Kariuki
Climate Domain Lead:	Mike Simion
Data and Governance Lead:	Gracious Ngetich
Partnership & User Engagement Lead:	Isaac Baya
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Requested Funding: US\$10,000 | Duration: 12 Months

Submitted to the UbuntuNet Alliance for Research and Education Networking
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1. Project Summary

Climate change in Kenya rarely arrives as a single shock. Households in flood-prone, poorly-served counties typically face **compound infrastructure disruption**, water supply failure, impassable roads, electricity outages and localised flooding occurring together, yet county governments still allocate adaptation resources using generic vulnerability maps rather than household-level evidence. Our team has already built and tested a working AI prototype that closes this gap. Using the nationally representative Kenya Housing Survey 2023/24 (21,347 households, 47 counties), we engineered a compound climate disruption target and trained an ensemble machine learning model (HistGradientBoosting) that predicts which households and counties face simultaneous climate-linked infrastructure failure, achieving a ROC-AUC of 0.881 (a 9.8-percentage-point improvement over a non-AI logistic regression baseline). We layered on SHAP explainability to make every prediction auditable, and a Data Envelopment Analysis (DEA-BCC) model that scores each county's adaptation capacity relative to its resources. The two are fused into a population-weighted **Urgency Index** and Policy Quadrant Matrix that tells a county officer, in one number, whether their county needs emergency funding, capacity mobilisation, preventative structural investment, or is a best-practice model to replicate. A working choropleth dashboard with an AI assistant has already been built on this output. This project will take that proof-of-concept (currently a research notebook and a static dashboard) and turn it into a deployed, user-tested decision-support tool piloted with two County Climate and Environment Departments and Kenya's National Disaster Management Authority (NDMA). At the end of 12 months we will have a live, county-facing web application, documented user feedback from a real government pilot, and evidence-based reprioritisation of adaptation resources reaching an estimated 21.3 million Kenyans identified as being at compound climate risk by 2026.

2. Problem Definition

Kenya's exposure to climate variability is well documented at the national level; what is not captured by that national picture is the *operational* problem county governments actually face on the ground. Climate risk here rarely arrives as one clean hazard. It arrives as **compounding infrastructure failure**: a single flood event can simultaneously cut a household off from piped water, wash out the only feeder road, and knock out the local electricity transformer. Our analysis of the Kenya Housing Survey (KHS) 2023/24 confirms this is the norm, not the exception, **35.5% of the 21,347 surveyed households** report experiencing at least one of water disruption, road access failure, electricity outage, or neighbourhood flooding, with electricity disruption alone affecting **31.6%** of households nationally. A problem that shows up this often, and this jointly, cannot be managed one hazard at a time.

Nor is it evenly spread. Our county-level analysis identifies four counties, Taita-Taveta, Kakamega, Busia and Kwale, at **critical urgency**, where high predicted household risk coincides with weak adaptation capacity, and a further thirteen counties at high urgency. Scaled against 2026 population projections (55.77 million citizens nationally, escalated from the 2019 KNBS census at a 2.3% CAGR), our model estimates that **21.3 million Kenyans (38.2% of the projected population)** live in households at elevated risk of compound climate disruption. Kakamega and Nairobi alone carry more than 1.5 million and 3.0 million at-risk citizens respectively, so this is simultaneously a rural infrastructure problem and an urban resilience problem, not one or the other.

The institutional gap here is not a lack of data. Kenya already runs a national household survey and holds a rich administrative dataset on water, roads and electricity; the gap is that this data has never been fused into a predictive, prioritisation-ready tool a county planner, KMD forecaster, or NDMA officer can actually use. County Integrated Development Plans and disaster response budgets are still allocated on historical incident reports and generic exposure maps, both reactive by design: they describe what has already gone wrong, not which specific wards and household clusters are statistically most likely to face compound shocks next, or which counties are under-performing relative to the adaptation resources they already have. Simple hazard overlays or single-indicator vulnerability indices, the realistic non-AI alternatives, cannot capture the *interaction* between physical hazard exposure, structural housing fragility, road isolation and economic strain that actually predicts compound disruption. This is not a hypothetical distinction: our own baseline comparison shows a standard logistic regression model underperforming our ensemble model by 9.8 points of ROC-AUC and 14.5 points of PR-AUC on the same task, on the same held-out households. Existing approaches are, in short, insufficiently granular, insufficiently explainable to non-technical county officers, and insufficiently linked to a governance-capacity measure that would let national government target scarce adaptation funding to where it is needed most and available least.

3. Existing Proof-of-Concept

3.1 What Currently Exists

Our team has already built and executed, end-to-end, a working AI prototype; not a plan for one. The artefact is a fully reproducible analytical pipeline (`AI_CLIMATE_RESILIENCE.ipynb`) plus a **live, deployed decision-support dashboard**, built on the nationally representative KHS 2023/24 microdata (21,347 households, 47 counties, 528 raw variables). The prototype spans the full analytical chain:

1. **Target engineering.** `compound_climate_disruption` is defined as any household reporting water disruption, road access failure, electricity outage, or flooding (35.5% national prevalence), with a strict anti-leakage rule excluding all four source columns from the feature set.
2. **A 34-feature predictive model** spanning hazard exposure, dwelling quality, utility access, demographics and county governance context, trained on a leakage-safe, stratified 70/15/15 split (14,942 / 3,202 / 3,203 households).
3. **A genuine non-AI baseline:** a class-weighted Logistic Regression, benchmarked against two AI ensembles (Random Forest, HistGradientBoosting) on the same held-out test set.
4. **Explainable AI:** SHAP attribution identifying the drivers of compound risk at population and household level.
5. **A County Climate Adaptation Capacity Index (CACI)**, via input-oriented Data Envelopment Analysis (DEA-BCC, PuLP), benchmarking each county's outcomes against its adaptation resources.
6. **An integrated Urgency Index and Policy Quadrant Matrix** fusing risk, vulnerability and capacity into one ranked, actionable output per county.
7. **A working, deployed front-end:** the *Kenya Climate Resilience Decision Engine*, a live choropleth dashboard (all 47 counties, KPI panel, sortable county table, and a Retrieval-Augmented Generation assistant grounded on the 47-county recommendation dataset, served via Groq and Gemini) publicly accessible at <https://kenya-ai-climate-resilience.vercel.app/>. The assistant answers county-specific questions on urgency tier, budget priority and intervention rationale by citing the underlyingly computed figures rather than general knowledge.

3.2 Evidence That It Works

This is not a conceptual pipeline: it has been run on real held-out data and produces validated, reproducible metrics.

Model	ROC-AUC	PR-AUC	Type
Logistic Regression (class-weighted)	0.782	0.667	Non-AI baseline
Random Forest (balanced subsample)	0.855	0.765	AI ensemble
HistGradientBoosting (best model)	0.881	0.812	AI ensemble

On the 3,203-household held-out test set, the AI ensemble delivers a **+0.098 ROC-AUC and +0.145 PR-AUC** improvement over the realistic non-AI alternative, direct, quantified evidence that AI materially improves predictive capability, as required by this call. The decision threshold was independently tuned on the validation set (0.512, F1 = 0.734) before being applied, untouched, to the test set, avoiding threshold leakage.

SHAP attribution confirms the model's drivers (electricity grid stability, road condition, dwelling exposure, and household income/dependency burden) are physically and institutionally sensible rather than spurious correlations, which matters directly for the transparency and auditability county officers will require before trusting the tool with resource-allocation decisions.

At the county level, the DEA-BCC model produced a full CACI ranking for all 47 counties, and the integrated Urgency Index identified **4 counties at critical urgency, 13 at high, 16 at moderate, and 14 at low**, with Taita-Taveta, Kakamega, Busia and Kwale ranking highest, concentrating an estimated 3.1 million citizens in the four most critical counties alone. This output is already exported to a structured county-level CSV and powers the live dashboard, demonstrating the pipeline runs cleanly from raw survey microdata to a decision-ready, public interface: the exact transition this call is designed to fund.

3.3 How the Live Application Works

The *Kenya Climate Resilience Decision Engine* is not a static export of model results; it is a working application with a deliberate architecture. The front end is a single-page interface (HTML/JS, Plotly.js) rendering a colour-coded choropleth of all 47 counties, switchable across five metrics (Urgency Score, AI-predicted compound risk, CACI adaptation capacity, observed disruption rate, and 2026 at-risk population), with a quadrant filter and a sortable county table so a user can move between a national overview and a single-county profile in two clicks. Selecting a county surfaces its full profile: urgency tier, quadrant classification, SHAP-derived risk drivers, and the recommended intervention, so the visual layer and the underlying model output are never separated.

The RAG assistant is built as **deterministic retrieval before generation**, not a raw LLM wired to a spreadsheet. A lightweight retrieval function first classifies the user's question, named-county lookup, urgency/priority ranking, capacity/CACI comparison, or a general query, and assembles only the relevant rows plus a national summary (population and risk totals, counts by urgency tier and quadrant, average CACI) before this context is passed to the language model. The model is explicitly instructed to answer only from the supplied county data, to cite the specific urgency score, at-risk population, CACI value or quadrant behind any recommendation, and to say plainly when a question cannot be answered from the data rather than guessing. This retrieval-first design is what lets a county officer trust a natural-language answer: every number the assistant states traces back to a specific row in the computed 47-county dataset, not to the language model's general knowledge.

3.4 Room for Improvement and Wider Adoption

We are transparent that the current build has real limits, and closing them is precisely the substance of the transition plan in Section 4. The retrieval layer is keyword-and-pattern based rather than

embedding-based, which works well at 47 rows but would need to move to vector retrieval if the tool is later extended to ward-level granularity. The dataset behind the dashboard is a static snapshot from a single KHS wave rather than a live-refreshing pipeline, and the assistant currently depends on a single LLM provider rather than a provider-agnostic fallback, both addressed in the pilot's hardening phase. Looking beyond Kenya, the underlying methodology, compound-shock target engineering, an ensemble classifier with a non-AI baseline comparison, SHAP attribution, and a DEA-BCC efficiency layer benchmarking outcomes against governance resources, is not Kenya-specific in its structure; it depends only on a household survey with comparable infrastructure-access questions and a set of administrative capacity indicators, both of which several UbuntuNet Alliance member countries already collect through their own national living-standards or housing surveys. This makes the pipeline a plausible template for other countries in the region, and is a concrete axis we intend to explore in the technical brief and case study this call asks us to produce.

3.5 Technology Readiness Level and Honest Gaps

By the TRL scale, we assess the current prototype at **TRL 5** (validated in a relevant environment: the full pipeline, i.e. target engineering, the 34-feature HistGradientBoosting model, SHAP attribution, and the DEA-BCC CACI, runs end-to-end on real KHS microdata and is already deployed as a public, functioning application with a working RAG assistant, not just a notebook). We are honest about what remains; these are precisely the gaps this grant will close:

- No county government, NDMA, or WASREB representative has yet used the live tool; target users are identified (Section 4) but a structured pilot with documented feedback has not yet run.
- The model uses a single cross-sectional survey wave (KHS 2023/24); it does not yet ingest real-time weather/hydrological signals or have a maintained data-refresh pipeline for future survey rounds.
- SHAP and CACI outputs exist in the dashboard but not yet as plain-language, per-county explanations tailored for a non-technical planning officer.

4. Proposed Transition from Proof-of-Concept to Pilot

4.1 AI Approach and Data Inputs

We will carry forward the validated modelling architecture from the proof-of-concept unchanged in substance (the 34-feature, leakage-audited HistGradientBoosting classifier with ROC-AUC 0.881, the SHAP attribution layer, and the DEA-BCC County Climate Adaptation Capacity Index), and invest the grant period in three things the PoC does not yet have: a live, user-facing application; real government users; and a documented pilot. Data inputs remain the existing KHS 2023/24 microdata and county administrative indicators (water system efficiency, EIA approval rates, planning staff counts, storage capacity) already used in the PoC, meaning **no new primary data collection is required**. This is a direct response to this call's preference for projects that build on datasets that already exist. Where county partners can supply updated administrative figures during the pilot (e.g. current water storage capacity or planning staff establishment), we will refresh the CACI inputs for their county specifically, giving pilot counties a more current picture than the notebook baseline.

4.2 From Notebook to Application

The transition path is deliberately narrow and de-risked, because the dashboard is already live, not merely built and waiting for deployment: the grant funds hardening and user validation of a real, functioning system, not construction from scratch. Concretely, we will:

1. **Harden the existing live dashboard.** The deployed Kenya Climate Resilience Decision Engine (choropleth map, KPI panel, sortable 47-county table, and the Groq/Gemini RAG assistant) will be extended into a **county-planner workflow view**: a per-county drill-down page showing the county's Urgency tier, Policy Quadrant classification, recommended intervention, SHAP-based plain-language risk drivers, and CACI peer-benchmarking against comparable counties.
2. **Add a role-appropriate explanation layer.** We will translate SHAP outputs (currently a technical beeswarm plot) into short, plain-language "why this county is flagged" summaries, so that a Climate and Environment Officer without a data science background can trust and act on the score without needing to interpret a model artefact directly, directly addressing the UNESCO transparency requirement in Table 1 of the call.
3. **Refresh and validate with pilot-county data.** We will engage two pilot counties selected from our current Critical/High urgency ranking (prioritising a rural county such as Kakamega or Busia and an urban county such as Nairobi or Mombasa, to test the tool across contrasting operating environments), incorporate any updated administrative figures they can share, and re-run the CACI and Urgency computation for those counties as a validation exercise.
4. **Run a bounded micro-pilot.** Over a defined testing window, County Climate and Environment Department staff and an NDMA liaison will use the dashboard to review current risk rankings against their own institutional knowledge, flag discrepancies, and complete a structured usability and trust survey, generating the credible evidence of usability and value this call explicitly asks for.

5. **Iterate on the interface and, where warranted, the model**, based on pilot feedback: for example, adding filters county officers request, or recalibrating thresholds if pilot users identify systematic false positives/negatives in their own local knowledge.
6. **Document and disseminate**, producing a short technical validation note and a public case-study brief usable for follow-on funding conversations with WASREB, NDMA, or county governments beyond the two pilot sites.

4.3 Innovation: Technical, Contextual and Practical

The technical novelty of this project is not any single algorithm; it is the **fusion of three distinct analytical layers into one decision** that a county officer can act on immediately: (i) a predictive layer (who is at risk, and why, via ML + SHAP); (ii) a governance-efficiency layer (which counties are underperforming relative to their own resources, via DEA-BCC); and (iii) a population-weighted prioritisation layer that combines both into a single, ranked, four-quadrant policy recommendation. To our knowledge, no existing Kenyan climate dashboard combines a validated compound-shock risk model with a formal efficiency-frontier governance benchmark in a single, county-facing tool. Contextual relevance comes from grounding every input in Kenya's own national household survey and administrative data, rather than global climate-exposure proxies that do not reflect local infrastructure realities. Practical applicability comes from the explicit design decision to require no real-time satellite feeds or IoT sensor networks: the tool works today, in data-constrained county offices, on data Kenya already collects.

4.4 Expected System Maturity After Project Completion

We expect the system to reach **TRL 6/7** by month 12: a technology demonstrated in a relevant, real operational environment (two county government pilots), with documented user feedback, rather than a laboratory-only artefact. Concretely, tangible and measurable outputs are:

- A publicly accessible, live decision-support web application covering all 47 counties.
- A documented pilot in at least two county government departments, with a structured usability/trust evaluation (target: at least 10 individual users across the two pilot sites, and a usability score benchmark established for future scale-up conversations).
- A plain-language, SHAP-derived explanation layer embedded directly in the interface for every county.
- A refreshed, validated CACI and Urgency ranking for the two pilot counties incorporating current administrative data.
- A public technical brief and case-study document suitable for follow-on funding applications (e.g. to WASREB, NDMA, or national climate finance mechanisms) and for submission as a peer-reviewed conference or working paper output from Strathmore University.

We are also explicit about the impact we are *not* claiming: this grant funds a bounded, two-county micro-pilot, not a national rollout, and it does not fund new primary data collection or continuous real-time monitoring. Those are natural, well-evidenced next stages that a successful pilot, with the credible usage evidence this call is designed to generate, would put us in a strong position to pursue with larger funders.

5. Implementation Plan and Timeline

Phase	Key Activities	Milestone / Responsible
Months 1–3	Harden existing dashboard for public deployment; build county drill-down and plain-language SHAP explanation views; identify and formally engage two pilot counties and an NDMA liaison; secure letters of collaboration.	Live dashboard deployed; two signed pilot-county engagements. <i>Technical Lead; Partnership & User Engagement Lead</i>
Months 4–6	Refresh CACI/Urgency inputs with pilot-county administrative data; onboard pilot users; run structured micro-pilot sessions and usability/trust survey; log discrepancies against local institutional knowledge.	Pilot sessions completed in both counties; usability data collected. <i>Principal Investigator; Climate Domain Lead</i>
Months 7–9	Iterate on interface and model calibration based on pilot feedback; strengthen data governance documentation; begin scoping integration with NDMA early-warning workflows.	Revised, pilot-informed version of the tool released. <i>Data and Governance Lead; Technical Lead</i>
Months 10–12	National dissemination (all 47 counties visible on the live dashboard); publish technical validation brief and case study; prepare peer-reviewed / conference submission; compile follow-on funding package.	Public brief published; follow-on funding proposal drafted. <i>Principal Investigator; Data Engineer</i>

Risk buffer: each phase reserves the final two weeks for catch-up, since pilot scheduling depends on county government availability outside our control (mitigation detailed in Section 8).

6. Project Team and Roles

Our team is deliberately structured to cover the full chain this call requires: technical AI delivery, geospatial and climate-domain grounding, data governance, and real user/partner engagement. We already have a track record of shipping a related tool together.

Name	Role	Contribution to This Project
David Mauti	Principal Investigator	Overall accountability, coordination across delivery phases, and primary liaison with Strathmore University and UbuntuNet Alliance on compliance and reporting. Research Scholar at iLabAfrica building dashboards and stakeholder-facing research briefs across multiple concurrent projects, with cross-sectoral background in insurance, public finance and agricultural data analysis.
Jakes Kariuki	Technical Lead	Owns the technical transition to deployed application: hardening the dashboard, extending the SHAP explanation layer, and integrating pilot-county data refreshes into the pipeline. Full-stack developer (Laravel, Vue.js, Next.js, FastAPI) and Research Scholar at iLabAfrica, where he currently builds AI capabilities, including a citizen chatbot and an agentic chat interface, into a live county revenue management system, directly transferable to hardening the county-facing dashboard and its RAG assistant.
Mike Simion	Climate Domain Lead	Grounds the project in Kenya's climate adaptation policy landscape; validates that risk drivers and recommended interventions are institutionally realistic; leads engagement content for county officers and NDMA. Co-author of a peer-reviewed paper modelling extreme rainfall events in Kenya using Extreme Value Theory (stationary/non-stationary GPD models with sea-surface-temperature and atmospheric-moisture covariates); prior industrial attachment in a County Finance and Economic Planning Department, including work on a County Integrated Development Plan.
Gracious Ngetich	Data and Governance Lead	Owns data quality, privacy and responsible-AI documentation; oversees the anti-leakage/provenance discipline already built into the PoC. Software developer with production experience building RAG-powered retrieval systems and AI-model integrations (Claude, Gemini) with structured error-handling and monitoring, directly relevant to hardening the dashboard's RAG assistant and its data-provenance logging.
Isaac Baya	Partnership & User Engagement Lead	Formally engages the two pilot counties and the NDMA liaison; secures letters of collaboration; designs and runs the structured usability/trust survey. Background in economics and statistics with field-research experience at the Kenya Industrial Research and Development Institute (stakeholder-facing feasibility studies and report writing); holds a Certificate in AI Leadership and Strategy covering risk management and ethical governance, relevant to Section 8.
Valerie Jerono	Data Engineer	Maintains the KHS 2023/24 pipeline (extraction, cleaning, feature engineering, anti-leakage audit); supports CACI/DEA-BCC computation and county data refresh. MSc Data Science candidate, Strathmore iLabAfrica; has built production full-stack applications and pipelines across other projects (including the Haraka platform and a County Rev-

6.1 Prior Collaboration and Readiness

The team's readiness is evidenced by the proof-of-concept itself: a leakage-audited, 34-feature ensemble model, a DEA-BCC governance layer, and a live public dashboard with a working RAG assistant already exist and were delivered by this team ahead of submission. Individually, team members bring directly transferable production experience: deployed RAG and LLM-integration systems (Kariuki, Ngetich), full-stack data pipelines and decision-support tools shipped across other production projects (Jerono), geospatial dashboards and automated GIS pipelines for municipal clients (Kiprono), and applied climate-risk statistical modelling published in a peer-reviewed venue (Simion). The pilot phase therefore extends skills the team has already exercised, rather than acquiring them from scratch. CVs, publication records and the partner letter of collaboration are included in Appendix A and Appendix B of this proposal.

7. Data and Infrastructure

Data sources. The core dataset is the Kenya Housing Survey 2023/24 (KHS), a nationally representative household survey administered by the Kenya National Bureau of Statistics (KNBS), which our team already holds and has processed (21,347 households, 47 counties). County-level administrative indicators used in the CACI model (EIA approval rates, water system efficiency, planning staff counts, water storage capacity) are drawn from existing public/administrative sources already integrated into the proof-of-concept. Population projections use the 2019 KNBS Census escalated to 2026 at a 2.3% CAGR. **No new primary household data collection is required for this project**, pilot counties will be asked only to confirm or update a small number of already-modelled administrative indicators specific to their county.

Access arrangements. The KHS 2023/24 microdata is already in our possession under existing academic access arrangements through Strathmore University; no new data-sharing agreement is required to continue using it. Any county-specific administrative data shared during the pilot will be provided directly by pilot-county officers under a data-sharing understanding documented in the letters of collaboration.

Data governance. All modelling work follows the anti-leakage and stratified-validation discipline already built into the proof-of-concept (target-defining columns strictly excluded from features; all preprocessing fitted on training data only). The Data and Governance Lead will maintain a data dictionary and provenance log for any pilot-county data ingested, and will document known limitations of the KHS sampling frame (e.g. potential under-representation of the most remote, hardest-to-reach households, who may in fact be at the highest compound-disruption risk) transparently in both the technical brief and the user-facing dashboard.

Technical infrastructure. The application layer is intentionally lightweight and low-cost, consistent with this call's preference against large infrastructure spend: a static front-end (HTML/JS, choropleth mapping) served from a standard cloud hosting platform, a small serverless function for the natural-language assistant, and no dedicated database or GPU infrastructure required. The modelling pipeline runs on standard Python data-science tooling (pandas, scikit-learn, SHAP, PuLP) and can be re-executed on any standard cloud or university compute resource. **Dependency on external systems** is limited to: (i) a third-party large-language-model API for the optional natural-language assistant feature (non-critical, the core map, KPI and table views function fully without it), and (ii) standard cloud hosting for the static site and serverless function. Both are low-cost, replaceable, and do not create lock-in risk to the core decision-support functionality.

8. Governance, Ethics and Risk Management

8.1 Risk Register

Risk	Description	Mitigation
Technical / model	Model or SHAP outputs do not generalise well to a specific pilot county's local context.	Re-validate against county officer knowledge at onboarding; treat disagreement as iteration data, not failure; limitations documented in-interface.
Data quality/access	County-supplied administrative data during the pilot is incomplete, delayed or inconsistent with KHS baselines.	Pilot proceeds on the validated KHS baseline regardless; county data is an enhancement, not a dependency; provenance log flags source of every figure.
Operational / adoption	Officer time and institutional coordination constraints slow engagement or usage during testing.	Written collaboration secured before Month 1; sessions scheduled around officers' calendars; two-week buffer built into every phase.
Deployment	Reliance on a third-party LLM API for the assistant feature introduces cost/availability risk.	Core dashboard (map, KPIs, table) works fully without the assistant; it is scoped as an enhancement on a low/no-cost tier.

8.2 Alignment with Governing Frameworks

Framework	How this project aligns
AU Continental AI Strategy	African-led and university-based, built entirely on Kenyan national data and county governance structures rather than imported proxies; strengthens a local research ecosystem (Strathmore iLabAfrica), builds a reusable local data asset (the KHS-derived county risk dataset), and is delivered by an interdisciplinary team.
UNESCO Ethics Recommendation	Human-centred design via a plain-language SHAP layer for non-technical officers; inclusiveness through documented KHS sampling limitations (Sec. 7) and deliberate inclusion of under-resourced counties (e.g. Taita-Taveta, Busia, Kwale) in pilot selection; data responsibility via the anti-leakage, train-only-fitting discipline already in the pipeline; transparency via a technical brief documenting assumptions and limitations.
EU AI Act	Classified as decision-support , not automated decision-making: the tool informs a human officer and does not autonomously allocate funding. We maintain basic technical documentation (model card, data dictionary, limitations note) and pair every score with its explanation, anticipating future transparency and human-oversight requirements without claiming full regulatory compliance.

Key ethical commitment. Because this tool ranks counties (and, implicitly, communities) by risk and adaptation capacity, we commit to never showing an Urgency score or CACI ranking without its explanation, and to caveating that the DEA-BCC “efficiency frontier” measures performance relative to existing resources, not absolute need, so a low-capacity county is read as under-resourced, not underperforming.

9. Communication

Results will be communicated through three channels aimed at three audiences. For **pilot-county officers and NDMA**, we will hold in-person or virtual walkthroughs of the live dashboard at pilot onboarding and again at Month 9, plus a one-page, plain-language county brief generated directly from the tool. For the **national policy and research community** (WASREB, Ministry of Environment, other county governments, and the UbuntuNet Alliance), we will publish a public technical brief and case-study at project close, present findings at the UbuntuNet Connect 2027 Conference showcase, and pursue a peer-reviewed or working-paper submission through Strathmore University. For **broader uptake**, the live dashboard itself remains publicly accessible after the grant period, and our team will use documented pilot evidence to approach follow-on funders (national climate finance mechanisms, WASREB, or philanthropic partners) for scale-up beyond the two pilot counties.

10. Budget and Justification

No.	Description	Number	Units	Unit Cost (US\$)	Total (US\$)
<i>Personnel Costs</i>					
1	Research assistant: data validation, pilot-county data refresh and provenance logging	1	12 months (part-time)	166.67	2,000
2	Technical development stipend: dashboard hardening, SHAP explanation layer, integration (Technical Lead)	1	6 months (part-time)	333.33	2,000
Sub-Total Personnel Costs					4,000
<i>Other Costs</i>					
3	Cloud hosting, serverless functions and LLM API costs (12 months)	12	months	41.67	500
4	Data pipeline and compute (validation runs, pilot-county recomputation)	1	lump sum	500	500
5	County pilot engagement: travel, venue and per-county coordination (2 counties)	2	counties	1,000	2,000
6	Community dissemination and training materials (pilot workshops + final brief design/print)	1	lump sum	1,500	1,500
7	UbuntuNet Connect 2027 showcase participation (materials/registration; travel excluded, covered separately by lead institution where applicable)	1	lump sum	500	500
8	Contingency (schedule/data-access risk buffer, Section 8)	1	lump sum	1,000	1,000
Sub-Total Other Costs					6,000
TOTAL BUDGET					10,000

Budget Notes. No institutional overheads are included, in line with call requirements. Line 1 funds a research assistant to carry out the data validation, county-data ingestion, and provenance logging described in Sections 4 and 7; this is the single largest determinant of pilot credibility and is prioritised accordingly. Line 2 is a modest development stipend, not a full salary, reflecting that the core model and dashboard already exist and this grant funds their hardening and extension, not construction from scratch. Line 3 keeps recurring infrastructure costs to a minimum, consistent with our lightweight, no-database architecture (Section 7); the LLM-assistant feature is scoped to fit within this line using a free/low-cost provider tier, and the core decision-support functionality does not depend on it. Line 5 is sized for two counties, matching our bounded micro-pilot design (Section 4) rather than a national

rollout. Line 8 provides a contingency directly tied to the two risks flagged in Section 8 (county officer scheduling delays; data-access variability) rather than being an undifferentiated buffer. We consider this budget proportionate: it funds the specific, named activities that take an already-working prototype to a documented, user-tested pilot, and nothing beyond that scope.

Appendix A: Curricula Vitae of Key Team Members

Prepared using the CV template in Appendix B of the Call for Proposals.

David Mauti: Principal Investigator

Institution: Strathmore University, Strathmore Institute of Mathematical Sciences (SIMS) **Country:** Kenya

Email: davidongaroh@gmail.com **Phone:** +254 101 772 348

Proposed role: Overall project leadership, coordination across delivery phases, and primary liaison with Strathmore University and the UbuntuNet Alliance on compliance and reporting.

Relevant expertise: Final-year MSc Data Science and Analytics candidate, Strathmore University; BSc Economics and Finance, Kenyatta University; Diploma in Software Engineering, Moringa School. Research Scholar, iLabAfrica (June 2025–present), conducting quantitative and qualitative analysis across multiple concurrent research projects, leading data collection and validation workflows, and drafting research briefs and stakeholder reports across non-specialist and technical audiences: directly transferable to PI-level coordination and UbuntuNet Alliance reporting. Prior roles spanning insurance (Customer Experience Officer, CIC Group), public finance (Junior Finance Officer, Uwezo Fund), and agricultural data (Data Assistant, Cereal Growers Association) give cross-sectoral grounding in the multi-stakeholder, data-constrained environments this project operates in. Core skills: Python (pandas, numpy, scipy), SQL, Power BI, survey design and data validation, dashboard development.

Selected relevant experience: Multiple active research projects at iLabAfrica (2025–present), building interactive dashboards and translating research findings for policymakers and non-specialist stakeholders; directly relevant to communicating Urgency Index and CACI outputs to county officers and NDMA.

Jakes Kariuki: Technical Lead

Institution: Strathmore University, iLabAfrica Research Centre **Country:** Kenya

Email: jakes.kariuki@strathmore.edu **Phone:** +254 791 736 258

Proposed role: Technical delivery of the dashboard hardening, SHAP explanation layer, and pilot-data integration described in Section 4.

Relevant expertise: MSc Data Science and Analytics candidate, Strathmore University, iLabAfrica Research Centre (2025–present), building on a BSc in Mathematics and Computer Science (Statistics Option), First Class Honours, Multimedia University of Kenya. Research Scholar at iLabAfrica (June 2025–present), currently building AI capabilities (a citizen-facing chatbot, an executive briefing service, and an agentic chat interface) into a live county revenue management system using SQL, Laravel and Vue.js. Core skills: Python, R, SQL, Scikit-learn, XGBoost/LightGBM/CatBoost, SHAP explainability, FastAPI, Laravel, Vue.js, Next.js, Docker, cloud deployment (Vercel, Azure Container Apps). Deployment experience spans multiple dashboards (Power BI, Streamlit, Next.js/Supabase) shipped end-to-end.

Selected relevant experience: AI Capabilities for a County Revenue Management System (iLabAfrica, 2025–present); Technical Lead role, contributed AI features (chatbot, executive briefing,

agentic interface) to a live government platform and resolved dashboard-to-report KPI discrepancies. Multi-Country Food Insecurity Classification Pipeline (independent); built a CatBoost/LightGBM/stacking-ensemble pipeline with SHAP-based cross-country feature importance on household survey data, directly analogous to the compound-disruption model in this proposal.

Mike Simion: Climate Domain Lead

Institution: Strathmore University, Strathmore Institute of Mathematical Sciences (SIMS) **Country:** Kenya

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Proposed role: Climate and policy grounding; validation of model features and recommended interventions; engagement content for county and NDMA users.

Relevant expertise: MSc Data Science and Analytics candidate, Strathmore University (SIMS); MSc Applied Statistics, Jomo Kenyatta University of Agriculture & Technology (2024–2026); BSc Applied Statistics with Computing, Maasai Mara University. Research Scholar, iLabAfrica (June 2025–present). Co-author of a 2026 peer-reviewed paper, *Modeling Extreme Rainfall Events Using Stationary and Non-stationary Peaks-Over-Threshold Frameworks in EVT*, developing Generalized Pareto Distribution models of Kenyan rainfall extremes using sea-surface-temperature and atmospheric-moisture covariates for climate risk assessment and disaster preparedness: directly relevant statistical grounding for validating the compound-disruption target and risk drivers in Section 3. Prior industrial attachment, Finance and Economic Planning Department, Narok County Government, contributing to the County Integrated Development Plan (CIDP) 2022–2027, giving first-hand experience with the county planning process this project’s outputs are designed to feed into.

Selected outputs: Simion Mike Mbumbu, Anthony Waititu, Herbert Imboga, Susan Mwelu (2026), published in the American Journal of Mathematical and Computer Modelling.

Gracious Ngetich: Data and Governance Lead

Institution: Strathmore University, Strathmore Institute of Mathematical Sciences (SIMS) **Country:** Kenya

Email: ngetich.gracious@protonmail.com

Proposed role: Data quality, privacy and responsible-AI documentation for the pilot; anti-leakage and provenance discipline oversight; hardening of the RAG assistant’s retrieval reliability.

Relevant expertise: MSc Data Science and Analytics candidate, Strathmore University (2027, in progress); BSc Business Information Technology, Moi University. Software Developer, Cygnite.Africa (Sept 2023–May 2026) and Data Science Researcher, Centre for Internet Governance and Policy Research, iLabAfrica (May 2025–present). Four years’ production experience architecting AI automation workflows (15+ deployed N8N workflows integrating Claude, Gemini and Replicate APIs), including RAG-powered retrieval systems for intelligent document synthesis, with structured error-handling, retry logic and monitoring for production reliability: a capability that maps directly onto hardening this project’s live Groq/Gemini RAG assistant and provenance logging. Certifications: AWS Certified Cloud Practitioner; Certified Data Engineer (Explore AI Academy).

Isaac Baya: Partnership & User Engagement Lead

Institution: Strathmore University, Strathmore Institute of Mathematical Sciences (SIMS) **Country:** Kenya

Email: isaacbaya92@gmail.com **Phone:** +254 728 981 837

Proposed role: Pilot-county and NDMA engagement, letters of collaboration, and designing and administering the usability and trust survey.

Relevant expertise: MSc Data Science and Analytics candidate, Strathmore University; BSc Economics & Statistics, Second Class Honours Upper Division, Maasai Mara University. Research & Data Scientist, iLabAfrica (May 2025–present), supporting innovation and startup-ecosystem research and translating datasets into stakeholder-facing dashboards and reports. Field research and stakeholder engagement experience at the Kenya Industrial Research and Development Institute (feasibility studies, investment facilitation, report writing to non-technical audiences). Certificate in AI Leadership and Strategy (AI Kenya, April 2026), covering risk management and ethical governance: directly relevant to the trust/usability survey design and Section 8 alignment work.

Valerie Jerono: Data Engineer

Institution: Strathmore University, iLabAfrica Research Centre **Country:** Kenya

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Proposed role: Maintains the KHS 2023/24 data pipeline (extraction, cleaning, feature engineering, anti-leakage audit); supports CACI/DEA-BCC computation and pilot-county data refresh.

Relevant expertise: MSc Data Science and Analytics candidate, Strathmore University, iLabAfrica Research Centre (May 2025–June 2027); BSc Actuarial Science, Daystar University (Aug 2019–Nov 2023). Research Scholar, iLabAfrica, building production full-stack applications and dashboards (Laravel, Vue.js, React) for government and healthcare clients, including MySQL/SQL pipeline and caching design for real transaction volumes. Has also contributed to the Haraka logistics platform and a County Revenue Management System, among other applied data and full-stack builds, giving broad, evidenced experience taking a data pipeline from notebook to deployed, decision-ready tool.

Links: linkedin.com/in/valerie-jerono github.com/VAL-Jerono

Patrick Kiprono: Geospatial AI Engineer

Institution: Strathmore University **Country:** Kenya

Email: patrickkiprono01@gmail.com **Phone:** +254 707 120 546

Proposed role: Owns all GIS, spatial and mapping components of the tool: the county choropleth layer, spatial joins between KHS households and county administrative boundaries, and spatial diagnostics during pilot-county data refresh.

Relevant expertise: MSc Data Science and Analytics candidate, Strathmore University (2025–present); BSc Geospatial Information Science with IT, Maseno University (2019–2023). Research Scholar and Machine Learning Instructor, Strathmore University (May 2025–present), delivering hands-on GIS and ML instruction and supervising data-driven research applying AI and geospatial intelligence across agriculture, healthcare and finance. Prior Geospatial Data Analyst, Digireg Ltd (Dec 2023–Sep 2025), automating geospatial workflows with Python and PostGIS for Dutch municipi-

pal tax and infrastructure monitoring, reducing reporting time by 40%, and building web dashboards visualising municipal land and property changes: directly transferable to the county choropleth and drill-down views this project extends. Core skills: Python, R, SQL, GeoPandas, QGIS/ArcGIS, PostgreSQL/PostGIS, Leaflet.js, Docker, FastAPI.

Selected relevant experience: Geospatial Analyst & Developer, Mapinfotek & Geomatiks Ltd (Jul–Dec 2023); applied GIS and remote sensing to map sugarcane farms and monitor crop growth using drone-based surveying, and automated data cleaning and validation workflows in Python, cutting manual effort by over 50%, evidencing the same spatial-data-to-decision pipeline this project requires at ward and subcounty level.

Links: linkedin.com/in/kipronopato github.com/Kipronopato

Appendix B: Supporting Materials

Letters of support/collaboration. This proposal builds on an existing institutional relationship: iLabAfrica’s County Revenue Management System (CRMS) and Own Source Revenue (OSR) Management System are already deployed with county government partners, giving the team a direct channel to pilot-county Climate and Environment Departments and NDMA for this project. Adoption of this compound-disruption decision-support tool alongside the CRMS/OSR systems helps counties not only understand their current climate-risk and revenue-capacity position in one place, but also opens the door to encompassing the two data streams together, so that risk and revenue data can eventually be mapped down to ward and subcounty level rather than remaining a county-wide aggregate. The letter of support or collaboration from the intended pilot-county partner or NDMA is appended here.

Proof-of-concept evidence. The following supporting evidence is available and should be attached or linked in the final submission package: (i) the executed analytical notebook (`AI_CLIMATE_RESILIENCE.ipynb`) showing all computed metrics reported in Section 3; (ii) the exported 47-county results file (`county_climate_recommendations.csv`); and (iii) the live, deployed Kenya Climate Resilience Decision Engine (`kenya-ai-climate-resilience.vercel.app`), including its Groq/Gemini-powered RAG assistant, as direct evidence that the pipeline runs end-to-end from raw survey microdata to a public, decision-ready interface.